

Lecture 62: BDBH204 - Machine Learning

Dataset of 10 emails

- Naive Bayes
- Prob graph model

Email	Offer x_1	Money x_2	Spam (c)
1	Y	Y	Y
2	Y	N	Y
3	Y	Y	Y
4	N	Y	Y
5	N	N	N
6	N	N	N
7	Y	N	N
8	N	Y	N
9	Y	Y	Y
10	N	N	N

Test Offer Y, Money Y prob that this is spam?

posterior
P

$$P(y|x) = \frac{P(x|y) P(y)}{P(x)}$$

$P(x)$

likelihood

prior

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = P(x_1, x_2 | y) P(y)$$

ignore denominator cuz it's constant

values are not really prob its more

Step 1: Calculate the higher probability like score

$$P(\text{spam}) = P(Y_{\text{offer}}) = \frac{5}{10} = \frac{1}{2}$$

$$P(\text{Not Spam}) = P(N_{\text{offer}}) = \frac{5}{10} = \frac{1}{2}$$

Step 2: Calculate likelihoods for spam class

$$P(\text{offer Yes} | \text{Spam}) = \frac{4}{5}$$

$$P(\text{money Yes} | \text{Spam}) = \frac{4}{5}$$

for not spam class

$$P(\text{offer Yes} | \text{Not Spam}) = \frac{1}{5}$$

$$P(\text{money Yes} | \text{Not Spam}) = \frac{1}{5}$$

$$P(\text{spam} | X) = P(X | \text{spam}) P(\text{spam}) \rightarrow \textcircled{1}$$

$$\begin{aligned} P(X | \text{spam}) &= P(x_1, x_2 | \text{spam}) \rightarrow \textcircled{2} \\ &= P(x_1 | \text{spam}) \cdot P(x_2 | \text{spam}) \end{aligned}$$

Substituting $\textcircled{2}$ in $\textcircled{1}$

$$\begin{aligned} P(\text{spam} | X) &= P(x_1 | \text{spam}) \cdot P(x_2 | \text{spam}) \cdot P(\text{spam}) \\ &= P(\text{offer Yes} | \text{Spam}) \cdot P(\text{money Yes} | \text{Spam}) \cdot P(\text{spam}) \\ &= \frac{4}{5} \times \frac{4}{5} \times \frac{1}{2} \\ &= \frac{8}{25} \end{aligned}$$

$$\begin{aligned} P(\text{Not Spam} | X) &= P(\text{offer Yes} | \text{Not spam}) \cdot P(\text{money Yes} | \text{Not Spam}) \cdot P(\text{Not Spam}) \\ &= \frac{1}{5} \times \frac{1}{5} \times \frac{1}{2} \\ &= \frac{1}{50} \end{aligned}$$

$$P(\text{spam} | x) = \frac{8}{25}$$

$$P(\text{Not Spam} | x) = \frac{1}{50}$$

Now add up to normalize

$$P(\text{spam} | x) = \frac{\frac{8}{25}}{\frac{8}{25} + \frac{1}{50}} = \frac{0.32}{0.32 + 0.02} = \frac{0.32}{0.34} = 0.941$$

$$= 94.11\%$$

$$P(\text{Not Spam} | x) = \frac{\frac{1}{50}}{\frac{8}{25} + \frac{1}{50}} = \frac{0.02}{0.32 + 0.02} = \frac{0.02}{0.34} = 0.058$$

$$= 5.89\%$$

posterior probability
calculation ✓